

Manas Mehta

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EDUCATION

GLS University

Bachelor of Technology in Computer Science & Engineering

Ahmedabad, Gujarat

Aug. 2023 – May 2027

- CGPA: 7.85/10 (Semester 6)
- Relevant Coursework: Deep Learning, Computer Vision, Data Structures & Algorithms, Linear Algebra, Database Management

EXPERIENCE

Independent Researcher

Jan. 2025 – Present

Self-Directed

- Conducting independent research on image captioning; proposed a novel SE-ResNet-50-D encoder with Squeeze-and-Excitation blocks paired with an LSTM decoder and soft attention for end-to-end caption generation
- Achieved BLEU-4 of 0.2301 and CIDEr of 0.6236 on MS-COCO Karpathy test split, outperforming Karpathy (2015) and Mao et al. (2014) baselines; manuscript under submission

ACHIEVEMENTS & COMPETITIONS

Smart India Hackathon (SIH) – National Finalist

Dec. 2025

Remote Sensing & Geospatial Domain

- Selected among top teams nationally for InfraNova, a Dual-Stream EDSR model for optical-guided thermal super-resolution; collaborated with a 4-member team to develop and present at national level

Meta × PyTorch OpenEnv Hackathon – Top 100 Finalist

Apr. 2026

Scaler School of Technology

- Selected in top 100 nationally; built TheSnitch, a scalable AI oversight environment training an LLM overseer to detect misbehavior in multi-agent tool-use traces using the OpenEnv framework

PROJECTS

InfraNova: Optical-Guided Thermal Super-Resolution | *PyTorch, Rasterio*

Aug. 2025 – Dec. 2025

- Engineered a Dual-Stream EDSR architecture fusing 10m optical and 30m thermal Landsat-8 imagery via ConvFuse + Channel Attention, achieving PSNR 42.4dB, SSIM 0.9269, and RMSE 0.0159
- Implemented two-phase training (decoder warmup → full fine-tuning) with combined L1 + SSIM loss; deployed live demo on Streamlit; refactored into modular Python package across 5 documented modules

Image Captioning: ResNet-D with SE Attention Blocks | *PyTorch* | [GitHub](#)

Jan. 2025 – Mar. 2026

- Trained with stabilized SCST using 80/20 mixed loss; achieved BLEU-4 of 0.2301 and CIDEr of 0.6236 on MS-COCO Karpathy test split, outperforming Karpathy (2015) and Mao et al. (2014) baselines; manuscript under submission

BERT Fake News Detector | *PyTorch, HuggingFace Transformers, Gradio* | [GitHub](#)

2025

- Fine-tuned RoBERTa-base on the LIAR dataset for binary fake news classification (REAL vs FAKE), achieving F1 of 0.7144 and Recall of 0.7827 on the test set
- Implemented full training pipeline with class-weighted loss to handle label imbalance; containerised with Docker and deployed Gradio demo on HuggingFace Spaces

TheSnitch: Scalable AI Oversight Environment | *PyTorch, OpenEnv, HuggingFace Spaces* | [GitHub](#)

Apr. 2026

- Designed an RL environment using OpenEnv to train an LLM overseer detecting misbehavior in multi-agent tool-use traces; implemented Trace-Outcome Consistency Check with held-out policy generalization
- Containerised and deployed on HuggingFace Spaces with Docker; submitted within 7-day competition window

TECHNICAL SKILLS

Languages: Python, SQL

DL Frameworks: PyTorch, HuggingFace Transformers, Scikit-learn, OpenCV

Libraries: NumPy, Pandas, Matplotlib, Rasterio

MLOps & Tools: Git, Docker, MLflow, Weights & Biases, Streamlit, FastAPI

Concepts: CNNs, Transformers, Attention Mechanisms, Transfer Learning, Super-Resolution, SCST, RLHF